Zhang 2019 Follicular Lymphoma — analysis summary

RUN 5 OF 6

GO / NO-GO GATES

Gate 1 — mixing & continuity → GO

Patients/samples intermix on one continuous manifold (iLISI 1.00→2.26 of 4 at Harmony θ=10; patient 1.67/2); single connected component, 0 detached islands. Strongly clonal FL batch integrates.

Gate 2 — module geography → GO (non-branching)

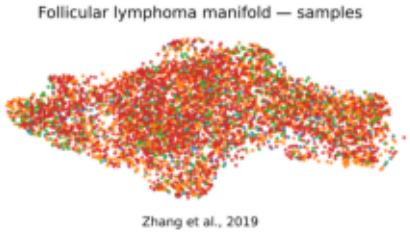
Non-branching: on the cycling-scoped Body, Gast/pSom co-vary (r=+0.33) and both oppose Crest (Crest↔Gast −0.83, Crest↔pSom −0.41); Crest is the terminal readout (high-decile pt=0.58). Single collapsed developmental axis.

Gate 3 — hub cross-repression → NA

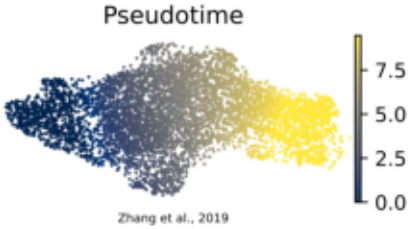
No Gast/pSom two-arm bifurcation (curve: 2 tips, 0 forks) → branching-only correlation/modules steps not produced.

STEP FIGURES

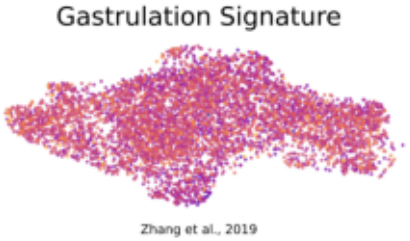
Manifold — samples



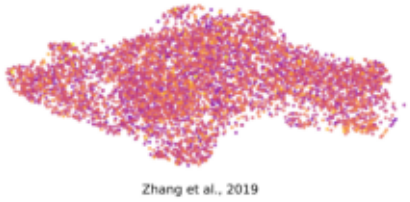
Pseudotime



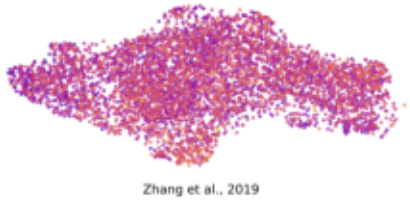
Gastrulation



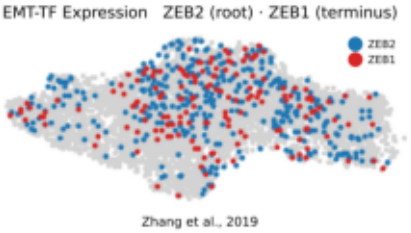
Somitogenesis



Neural Crest



EMT-TF expression



Dataset.
Follicular lymphoma (Zhang et al., 2019), 10x — 5,645 malignant B cells (trajectory scoped to 5,449 non-cycling).

Verdict.
Pattern reproduces (non-branching). A single Gast → Crest developmental axis: gastrulation/somitogenesis programs at the root, neural-crest as the terminal readout. No competing arms → Gate 3 not applicable. Gate 1 GO / Gate 2 GO / Gate 3 NA. Module means (raw) Gast +0.012 / pSom +0.005 / Crest +0.010. EMT-TFs ZEB2 (28%, earlier) → ZEB1 (12%, most terminal).

Parameter choices

Sample threshold	≥100 floor — kept all 4 samples (largest gap ratio 2.74, not a clean cliff; min 680 a genuine primary FL tumour)	**Integration**	Harmony θ=10 on sample (iLISI sweep + embedding; FL clones patient-private, need strong θ)
Embedding	n_pcs=20, n_neighbors=30, spread=1.3; 636 overdispersed genes	**Cycling scoping**	excluded a 196-cell, 95%-cycling lobe where pSom was proliferation-confounded (pSom↔cycling 0.61→0.18)
Geometry	Linear/non-branching scFates curve (ElPiGraph) N=20 λ=0.3 μ=100; 2 tips/0 forks; fold-free (SA 2.96, DC1 +0.94)	**Root**	Gast/pSom-high, Crest-low origin (Gast ρ=−0.41, Crest ρ=+0.43 along pt)
Module means	Gast +0.012 / pSom +0.005 / Crest +0.010 (raw)	**EMT-TFs**	ZEB2 (28%, earlier) · ZEB1 (12%, terminal); SNAI/TWIST/FOXC/PRRX <3%; VIM excluded